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Adaptive Machine Learning Algorithms for Data Processing in Transportation Systems

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Abstract

Urban transportation systems increasingly demand responsive and scalable predictive frameworks to manage fluctuating passenger volumes in real time. This study proposes a novel adaptive learning approach for short-term passenger flow forecasting in Astana's bus network, integrating lightweight models with online retraining mechanisms. By incorporating model updating strategies triggered by real-time data drift detection, the proposed framework adapts dynamically to temporal and structural changes in urban mobility patterns. A modular data pipeline was developed to preprocess transactional transport data, extract relevant features, and implement prediction routines using Adaptive Light Gradient Boosting and Incremental Forest ensembles. The proposed LightGBM-based system demonstrated reliable accuracy and stable predictive performance, ensuring a balance between responsiveness and model robustness. Although alternative models such as GBR demonstrated lower isolated error, LightGBM was selected due to its reproducibility and consistent behavior under real-time constraints. Compared to baseline models, our adaptive scheme significantly reduces lag in prediction correction and enhances robustness against anomalies. These

findings highlight the importance of continual learning paradigms in smart city mobility infrastructure, offering a practical solution for real-time transit optimization. © 2025 IEEE.

Author keywords

adaptive learning; incremental models; passenger flow forecasting; public transportation; real-time prediction; urban mobility

Indexed keywords

Engineering controlled terms

Behavioral research; Bus transportation; Data handling; Forecasting; Learning algorithms; Learning systems; Machine learning; Mass transportation; Smart city

Engineering uncontrolled terms

Adaptive learning; Adaptive machine learning; Flow forecasting; Incremental models; Passenger flow forecasting; Passenger flows; Public transportation; Real-time; Real-time prediction; Urban mobility

Engineering main heading

Urban transportation

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